

Modeling crop yield using a modified harvest index-based approach: application in chickpea

A. Soltani*, B. Torabi, H. Zarei

*Department of Agronomy and Plant Breeding, Gorgan University of Agricultural Sciences,
P.O. Box 386, Gorgan, Iran*

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Abstract

The linearity of harvest index (HI) increase has been used as a simple means to analyze and predict crop yield in experimental and simulation studies. It has been shown that this approach may introduce significant error in grain yield predictions when applied to diverse environments. This error has been ascribed to variability in the rate of linear increase in HI with time (dHI/dt). Data from two field experiments indicated in chickpea (*Cicer arietinum* L.) that dHI/dt varied among sowing dates. This variation was related to the length of pre-seed growth phase and vegetative growth (dry matter production) during this phase and could be described by the mean daily temperature from sowing to beginning seed growth. dHI/dt increased linearly with increase in the temperature up to 17 °C when it reached to its maximum value and remained constant. Simulation of these field experiments using a chickpea crop model including a constant dHI/dt resulted in yield over-prediction for some sowing dates. However, a modified HI-based approach greatly improved model predictions. In this approach, potential seed growth rate (SGR) is calculated using the linear HI concept, but actual SGR is limited to current biomass production and the remobilisation of dry matter accumulated in vegetative organs before the seed growth period. This modification well accounted for temperature and drought effects on HI and resulted in better yield predictions under conditions of major chickpea producing areas of Iran. Therefore, we recommend that the modification to be applied in the other HI-based models.

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1. Introduction

Many crop models predict seed yield from estimates of the number of seeds per unit area and the rate and duration of seed filling (e.g. Ritchie et al., 1998). Some other models use partitioning coefficients to estimate seed yield accumulated (e.g. Goudriaan and

* Corresponding author. Tel. +98 171 4420438;
fax: +98 171 4420981.

E-mail address: asoltani_99@yahoo.com (A. Soltani).

van Laar, 1994). In some other crop models, seed yield accumulation is calculated as the product of biomass accumulation and harvest index (HI), and harvest index is assumed to increase linearly as a function of time after beginning seed growth with a constant rate (dHI/dt) (Sinclair, 1986).

The last method is based on the concept developed in soybean by Speath and Sinclair (1985) that HI linearly increases with time over much of the seed growth period. This response has also been observed in sorghum and maize (Muchow, 1988), peanut (Bennett et al., 1993), wheat (Moot et al., 1996), sunflower (Bange et al., 1998), chickpea (Soltani et al., 1999), field pea (Lecoeur and Sinclair, 2001), pigeonpea (Robertson et al., 2001a) and fababeen (Turpin et al., 2002). Furthermore, it has been shown that dHI/dt remains stable over a range of growth conditions such as variations in sowing date, irrigation treatments and N level (Moot et al., 1996; Bindi et al., 1999; Lecoeur and Sinclair, 2001). The strength of this approach lies in its simplicity, intrinsically combining the contribution of current and stored assimilate to seed yield and so removing the need for complex predictions of seed number and size in the prediction of seed yield (Chapman et al., 1993). Use of a constant dHI/dt has proved effective and robust in a number of crop simulation models including soybean (Sinclair, 1986), maize (Muchow et al., 1990), wheat (Amir and Sinclair, 1991), sunflower (Chapman et al., 1993), and chickpea (Soltani et al., 1999).

However, dHI/dt is not constant under vast conditions. In the conditions of severe water limitation during seed filling, HI ceases to increase ($dHI/dt = 0$), e.g. Muchow (1989) and Siddique and Sedgley (1986). Hammer et al. (1995) found that dHI/dt varied among experiments in peanut. They related this variation to the average temperature from sowing to the end of leaf growth. Bange et al. (1998) in sunflower showed that while rate of increase in HI was linear, the rate decreased with low temperature during seed growth period. Hammer and Broad (2003) in sorghum found that dHI/dt was reduced considerably under cool conditions. Previously, Hammer and Muchow (1994) had shown that the lower precision in predicting grain yield in their sorghum model was associated with limitations in predicting HI.

Chickpea (*Cicer arietinum* L.) is one of the major pulse crops in the West Asia and North Africa

(WANA) region. It is cultivated on a large scale in arid and semiarid environments, and has considerable importance as food, feed and fodder (Saxena et al., 1996). In Iran chickpea is the most important legume crop, and is sown on more than 50% of the total legume area. In chickpea model of Soltani et al. (1999) seed yield is predicted using linear HI concept. Recently, when this model was used to simulate a few field experiments under conditions of Gorgan, some unreal HIs and hence grain yields were predicted (see Section 3.2). Gorgan is a temperate, sub-humid environment with mild winters where chickpea is sown in autumn or early winter and has a long vegetative growth period (3–6 months) before seed filling. This was the reason that we initiated the present study.

There were three objectives of this study: (1) to report relationship between dHI/dt and average temperature before seed growth period, (2) to evaluate a modified HI-based approach for predicting seed yield of chickpea under special conditions of Gorgan, and (3) to re-evaluate chickpea model of Soltani et al. (1999) containing the modified HI-based method. The modified HI-based approach is described in Section 2.2.

2. Materials and methods

2.1. Field experiments

Two field experiments were conducted at the Gorgan University of Agricultural Sciences Research Farm, Gorgan (36°85'N, 54°27'E and 100 m asl), Iran. Monthly climatic data of Gorgan is presented in Soltani et al. (1999). Total rainfall and average temperature during usual chickpea growing period (December–May) are 300 mm and 12 °C, respectively. The field experiments were not primarily conducted for the objectives of the present study. First experiment aimed at the study of phenological development and the second experiment aimed at the study of dry matter production and partitioning. The soil was a deep silty clay (fine-silty, mixed, active, thermic, Typic Calcixerolls) and both experiments were conducted under well-watered conditions.

First experiment (exp. 1) was a serially sown experiment that started in December 2001 and continued until August 2003. Four chickpea genotypes

(Beauvanij, 9096c, Hashem and Jam) were sown at 11 sowing dates. The design was a single split plot with sowing dates in the main plot and genotypes in the sub plot, replicated four times. Plot size was 1.5 m × 4 m. Plots were hand-seeded using row spacing of 25 cm. Target plant density was 50 plant m⁻² and two seeds were planted at correct spacing at 5 cm depth, and seedlings thinned to one in each position later. During the experiment weeds were hand-controlled and several sprayings were carried out against *Ascochyta* blight. At the end, there were nine sowing dates for cv. Hashem, the local cultivar. In this paper we have only used HI and yield data of Hashem at sowing dates (day of year, DOY) of 12 December (346) 2001 and 15 January (15), 13 February (44), 17 March (76), 16 April (106), 18 May (138), 15 September (258), 14 October (287) and 12 November (316) 2002.

Stages of development of emergence (50% of plants with some parts at soil surface), flowering (50% of plants with one flower at any node, R1), pod initiation (50% of plants with 0.5 cm pod at one of the four upper nodes with unrolled leaf, R3), pod filling (50% of plants with peas beginning to develop, R5) and maturity (50% of plants with pods yellowing, R7) were recorded every 2 days. At maturity, harvest index, grain yield and yield components were determined. To obtain dHI/dt, HI was divided by duration (in days) from beginning of linear HI increase to its cessation. Beginning and cessation of linear HI increase were taken as 0.27 and 0.95 of thermal time from R1 to R7, respectively, based on results of the second field experiment.

The second experiment (exp. 2) was conducted in growing season of 2002–2003. Factorial combinations of three sowing dates and four plant densities were the treatments of the experiment. The experimental design was a randomized complete block design with four replicates. Sowing dates (DOY) were 5 January (5), 6 March (65) and 28 April (118) of 2003 and four plant densities were 15, 30, 45 and 60 plant m⁻². Phenological stages, leaf area index (LAI), above-ground biomass (separated as leaf, stem plus pod wall, seed and dead material) were monitored and determined during the crop cycle. The number of sampling for LAI and above-ground biomass were 11–13 and for HI determination were 8–11. Samples were oven-dried at 70 °C to constant weight, and then weighed. HI was determined based on above-ground plant mass as the

ratio of seed weight to total weight of the harvested material. HI data were regressed against time (days after R1) with a three-piece segmented, non-linear regression model (Eq. (1)) (Soltani et al., 2004). This model separates HI increase into distinct phases; zero HI in phase 1; linear increase of HI in phase 2 and maximum HI in phase 3. The segmented model can be expressed as:

$$\begin{cases} y = 0, & \text{for } x < s \\ y = b(x - s), & \text{for } s < x < m \\ y = b(m - s), & \text{for } x > m \end{cases} \quad (1)$$

where y is the harvest index (%), x the time after R1 (days), b the rate of linear increase in HI (dHI/dt, %day⁻¹), s the time of beginning linear increase in HI, and m the time of cessation of the linear increase in HI. Fig. 1 indicates an example fitting of the regression model to data from plant density of 45 plant m⁻² at sowing date of 6 March in exp. 2. Statistical analysis was conducted using the Statistical Analysis System (SAS Institute, 1989).

In both experiments, daily maximum and minimum temperatures, sunshine hours and rainfall were measured at a standard weather station located a few meters from the experimental units.

2.2. The crop model used

The chickpea crop model of Soltani et al. (1999) was used in this study. This model is similar to the

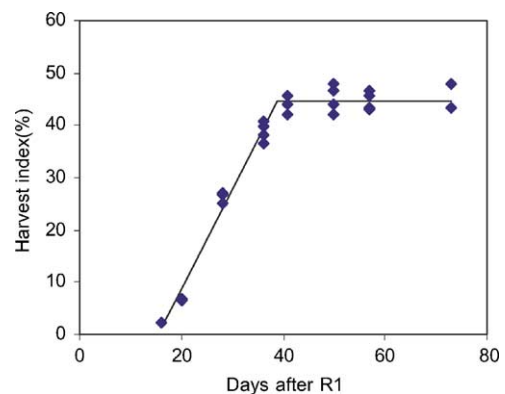


Fig. 1. Example fit of a segmented, non-linear regression model to harvest index data of chickpea vs. time after flowering (R1). Data from sowing date of 6 March and plant density of 45 plant m⁻² in exp. 2.

models of Sinclair (1986) and Hammer et al. (1995). The model is based on a daily time step and simulates crop growth and development as a function of temperature, solar radiation and water availability. Crop phenology is divided into two growth stages (before and after beginning seed growth), the duration of which is predicted based on photoperiod, daily temperature and water deficit. Leaf area development is calculated as functions of expansion and senescence of leaves. These functions are sensitive to temperature and water deficit. Daily biomass production is predicted from leaf area index, light extinction coefficient and radiation use efficiency (RUE). RUE is dependent on the average daily temperature (Soltani et al., 2001a) and soil water deficit (Soltani et al., 2000b). Transpiration is calculated as a function of daily biomass production, transpiration efficiency coefficient and daily vapor pressure deficit. Daily biomass production and leaf growth decrease below potential values once the fraction of transpirable soil water (FTSW) declines below a threshold value. Soil evaporation, soil water drainage, and runoff are also calculated in the soil water balance sub-model. The model uses readily available weather and soil information. At this stage the model does not account for the effects of pests, diseases and soil fertility. For details refer to Soltani et al. (1999, 2000b, 2001b).

In the original model (Soltani et al., 1999), seed growth rate for each day (SGR, $\text{g m}^{-2} \text{day}^{-1}$) is calculated as:

$$\text{HI}_t = \text{HI}_{t-1} + \frac{d\text{HI}}{dt} \quad (2)$$

$$\text{SGR} = \text{VDM}_{t-1} \times \left(\frac{\text{HI}_t}{1 - \text{HI}_t} \right) - \text{GDM}_{t-1} \quad (3)$$

$$\text{GDM}_t = \text{GDM}_{t-1} + \text{SGR} \quad (4)$$

$$\text{VDM}_t = \text{VDM}_{t-1} + \text{DBP} - (\text{SGR} \times \text{GCF}) \quad (5)$$

$$\text{TDM}_t = \text{VDM}_t + \text{GDM}_t \quad (6)$$

where VDM is vegetative dry matter, GDM the grain dry matter, TDM the total dry matter, DBP the daily biomass production, and GCF the grain conversion factor which allow for the energy content difference between the two types of tissue (VDM and GDM). SGR is calculated with an hourly time step which eliminates the numerical oscillations observed with the daily time step (Sinclair, 1986). Under severe water limitation during grain filling of chickpea HI ceased to increase (Siddique and Sedgley, 1986).

Owing to the lack of the data to quantify this effect, it is modeled by stopping the linear increase in HI if FTSW is below 0.1 and HI exceeds 0.2 as outlined by Chapman et al. (1993), Hammer et al. (1995) and Hammer and Muchow (1994). The minimum value for HI of 0.2 represents the maximum remobilization of stored assimilates. Any seed growth after this combination of conditions is limited to current biomass production. This correction significantly improved model predictions under rainfed conditions of Iran with terminal drought stress.

We modified the HI approach in the original model and prepared a modified model. In the modified model, every thing is the same as original model, but SGR as calculated above is considered potential SGR or grain demand for assimilates. The actual SGR (ASGR) is limited to the current biomass production (DBP) and the remobilization of dry matter (REMOB) accumulated in vegetative organs before seed growth period:

$$\text{ASGR} = \min(\text{SGR}, \text{DBP} + \text{REMOB}) \quad (7)$$

It was assumed that 17% of the total above-ground biomass at beginning seed growth can be re-mobilized to the grains during seed growth period. The remobilization percentage of 17% is also supported by the results of exp. 2 (data not shown). More experiments, however, are necessary to document the dependency of the percentage to crop biomass and the environment. The remobilization percentage is an input parameter and can be changed if required. Daily potential rate of remobilization depends on the ratio of daily thermal time (DTT) to the thermal time from beginning seed growth to the end of seed growth (TTLHI) as:

$$\text{REMOB} = \left(\frac{\text{DTT}}{\text{TTLHI}} \right) \times (\text{BLHIDM} \times 0.17) \quad (8)$$

where BLHIDM is the total above-ground biomass at beginning seed growth. Actual rate of remobilization is determined by seed demand for assimilate. Soltani et al. (2000a) in their chickpea model and Robertson et al. (2001b) in their pigeonpea model have used similar approach, but they did not evaluate the performance of the approach compared to the original one. In the modified model, there is no formula to describe cessation of HI increase under severe water limitation conditions as it is in original model.

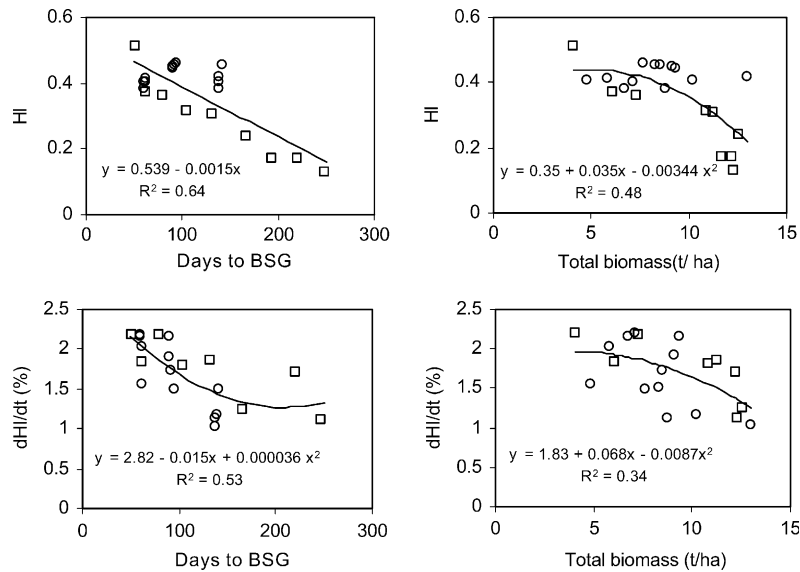


Fig. 2. Dependency of harvest index (HI) and dHI/dt to days to beginning seed growth (BSG) and total biomass across exp.1 (circles) and exp. 2 (squares).

In the original and modified models, dHI/dt was set at a constant value of 0.02 for cv. Hashem. This value is the maximum dHI/dt found for cv. Hashem (see Section 3.1, Fig. 3) and does not change by growth conditions such as temperature, solar radiation, crop density or sowing date. dHI/dt is a cultivar specific parameter. The function presented in Fig. 3 was not

included in the original or modified models. In another version of original model, dHI/dt value defined as a function of mean temperature from sowing to beginning seed growth. Hereafter, we use the symbol F3E for the original model that contains dHI/dt as a function of the temperature. The calendar date of beginning linear increase in HI (s) and its cessation (m) are

Table 1

HI-parameter estimates for chickpea cv. Hashem at four plant densities and three sowing dates of exp. 2

Sowing density (plant m ⁻²)	dHI/dt ± S.E.	s ± S.E.	m ± S.E.	Yield ± S.E.
5 January				
15	1.50 ± 0.14	13.7 ± 1.2	43.9 ± 1.8	3.29 ± 0.31
30	1.13 ± 0.11	10.3 ± 1.1	44.1 ± 2.0	3.79 ± 0.30
45	1.17 ± 0.10	11.2 ± 0.4	45.9 ± 1.9	3.61 ± 0.06
60	1.13 ± 0.11	10.7 ± 0.7	47.6 ± 2.9	3.37 ± 0.23
6 March				
15	1.49 ± 0.21	14.9 ± 2.9	45.8 ± 0.9	3.25 ± 0.38
30	1.73 ± 0.23	16.9 ± 2.3	43.2 ± 1.9	2.51 ± 0.06
45	1.91 ± 0.20	15.3 ± 0.7	38.8 ± 1.3	2.80 ± 0.10
60	2.15 ± 0.20	16.4 ± 0.6	37.1 ± 1.4	2.40 ± 0.31
28 April				
15	1.55 ± 0.30	11.9 ± 1.4	38.0 ± 1.5	1.93 ± 0.19
30	2.03 ± 0.14	13.3 ± 0.6	33.7 ± 0.5	1.72 ± 0.08
45	2.15 ± 0.21	12.7 ± 0.6	30.5 ± 0.6	2.08 ± 0.24
60	2.19 ± 0.33	12.3 ± 0.7	30.6 ± 1.1	1.89 ± 0.10

Parameter estimates are the linear increase in harvest index (dHI/dt, % day⁻¹), the time of beginning (s , day) and cessation (m , day) of linear increase in HI. Grain yields (t ha⁻¹) are also included.

predicted based on thermal time after flowering. These parameters are cultivar specific and are independent of management input such as crop density and sowing date. For cv. Hashem s and m are occurred 201 and 709 °C after flowering (see Section 3.1).

2.3. Model evaluation

Model evaluation was initially conducted for cv. Hashem at Gorgan by comparing simulated grain yield of original and modified models with those observed in the field experiments described above. Required parameters of cv. Hashem related to LAI development, phenology and dHI/dt were derived from the mentioned field experiments. Hashem has a longer life cycle, and higher maximum plant leaf area and dHI/dt compared to Jam. Therefore, data sets of the experiments are not entirely independent for model evaluation at Gorgan and cv. Hashem. However, the data of the experiments had not been used to calibrate the performance of the model.

Subsequently, the capability of the modified model to simulate crop yield was evaluated using the same data that were used for model testing by Soltani et al. (1999) (their Fig. 6d). The evaluation was carried out separately for water-limited (rainfed) conditions and irrigated conditions. All inputs related to chickpea cultivars (Jam and Kaka) were the same as Soltani et al. (1999). The value of dHI/dt for cv. Jam and Kaka in the original and modified model was 0.013. To simulate these field experiments, weather data, soil characteristics and the relevant management inputs that had gathered for each experiment were used.

Any difference in model performance between original and modified models is due to the modified HI-based approach that is used for modeling yield formation. Model performance was quantified by calculating root mean square of deviations (RMSD) and by regressing simulated yields versus observed yields.

3. Results and discussion

3.1. Field experiments

Fitting Eq. (1) to data of exp. 1 to obtain simultaneous estimation of HI-parameters using least square method was impossible. This equation was then fitted

to HI data versus time after flowering in exp. 2. Estimates of the HI-parameters resulted from fitting Eq. (1) to the data of exp. 2 are presented in Table 1. dHI/dt ranged between 1.13 and 2.19% day⁻¹. Sowing date differences for dHI/dt were significant, but at each sowing date (except for sowing date of 6 January) the differences between plant densities were not significant. At sowing date of 6 January, significant difference was found between plant density of 15 plant m⁻² and other plant densities for dHI/dt . The value of dHI/dt was similar (1.5% day⁻¹) for plant density of 15 plant m⁻² at all sowing dates (Table 1). The differences for the time of beginning and cessation of linear HI (in days) between plant densities at each sowing date were not significant, but were significant between sowing dates. They occurred sooner with delay in sowing date (warmer temperature). Fitting Eq. (1) to data of HI versus thermal time after R1 showed that the time of beginning and cessation of linear HI occurred after 0.27 and 0.95 of the thermal time from R1 to R7. A thermal time of 746 ± 24 °C day was obtained from R1 to R7 with a base temperature of 0 °C.

Calculation of dHI/dt for cv. Hashem at different sowing dates of exp. 1 also showed that this parameter was not constant. Across the both experiments, HI and dHI/dt were depended on duration of pre-seed growth phase and total biomass (Fig. 2). All the relationships were significant at $P \leq 0.05$. Duration of pre-seed growth phase accounted for 64% and 53% of the

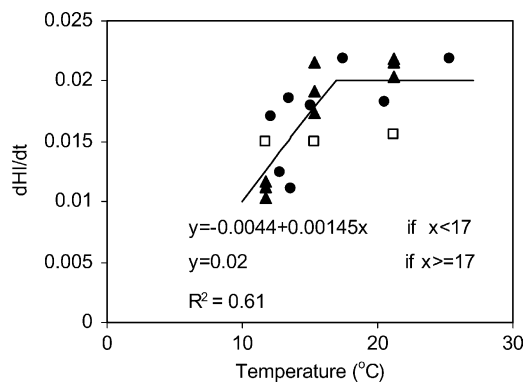


Fig. 3. Linear increase in harvest index (dHI/dt) of chickpea vs. mean daily temperature from sowing to beginning seed growth for exp. 1 (filled circles) and exp. 2 (filled triangle) at Gorgan. Open squares are from low plant density (15 plant m⁻²) in exp. 2. All the data have been used in line fitting.

changes in HI and dHI/dt, respectively. The greater duration of the phase and total biomass was related to lower HI and dHI/dt. These relations indicate the necessity for adjusting of dHI/dt based on pre-seed growth conditions experienced by the crop. Because duration of pre-seed growth phase and biomass are variable and depend on growing conditions, we examined the data for another more stable relationship for dHI/dt.

Across the both experiments, dHI/dt was depended on mean temperature from sowing to beginning seed growth (Fig. 3). The mean temperature before seed growth period accounted for 61% of dHI/dt variation. dHI/dt increased linearly with increase in the temperature up to 17 °C when it reached to its maximum value and remained constant. This relationship between dHI/dt and temperature is expectable because lower temperatures during this period result in longer the duration and the crop will have more time to accumulate biomass (Fig. 4). The maximum dHI/dt (0.02 or 2%) was included in the original and modified models for cv. Hashem. The function presented in Fig. 3 was not included in the original or modified models. However, F3E model contained the function. The dependency of dHI/dt to temperature is in agreement with Hammer et al. (1995). They found that dHI/dt declined rapidly when average temperature from sowing to the end of leaf growth exceeded about 27 °C. In

our data, there was no relationship between dHI/dt and temperature during seed growth period (data not shown). However, Bange et al. (1998) reported that the dHI/dt was not constant at very low temperatures (down to 9 °C) during grain filling, due to limitations to biomass accumulation and remobilization. Similarly, Hammer and Broad (2003) reported in sorghum that HI dynamics varied among experiments. The average dHI/dt was reduced considerably in one experiment in cool condition.

3.2. Simulation of the field experiments

In exp. 1 days to flowering varied between 44 and 232, days to maturity between 66 and 271, and grain yield between 1.60 and 3.45 t ha⁻¹ (Table 2). Simulation of exp. 1 using original and F3E models is presented in Fig. 5. Original model gave acceptable yield prediction for sowing dates between DOY 15 and 138. Conventional sowing date of chickpea at the location is between DOY 350–365 and 1–70. However, original model largely over-predicted grain yields for sowing dates between 258 and 346. At these sowing dates, chickpea crops have a long and excessive vegetative growth. RMSD of original model was 1.14 t ha⁻¹ (46% of observed mean) and there was a correlation coefficient of 0.46 between observed and simulated yields (Fig. 5). Thus, using a constant dHI/

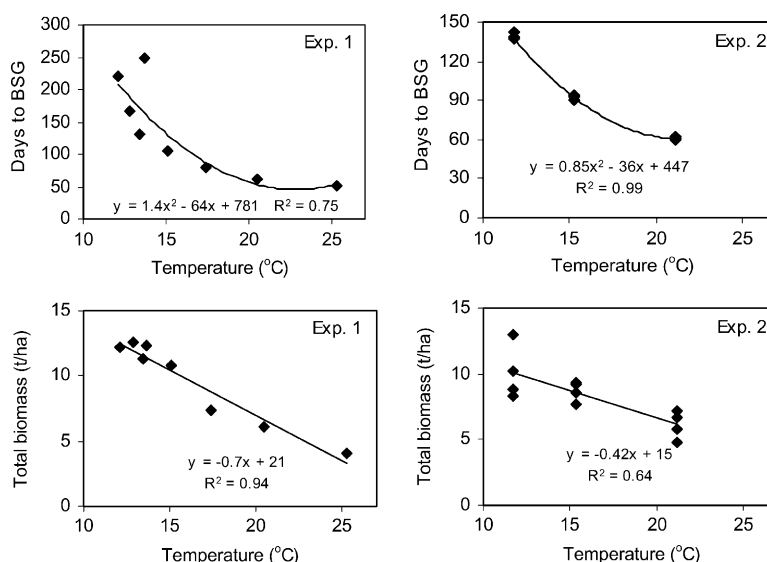


Fig. 4. Dependency of days to beginning seed growth (BSG) and total biomass to mean temperature from sowing to BSG in exp.1 and exp. 2.

Table 2

Days to flowering (DTF), mean temperature (TMPF, °C) and solar radiation (SRF, MJ m⁻² day⁻¹) from sowing to flowering, days to maturity (DTM), mean temperature (TMPM, °C) and solar radiation (SRM, MJ m⁻² day⁻¹) from flowering to maturity, and grain yield (YLD, t ha⁻¹) for cv. Hahsem at different sowing dates in exp. 1

Sowing date (DOY)	DTF	TMPF	SRF	DTM	TMPM	SRM	YLD
12 December 2001 (346)	145 d	9.8	10.3	183 d	22.2	20.4	3.02 b,c
15 January 2002 (15)	119 e	12.8	13.1	150 e	23.5	22.2	3.45 a
13 February 2002 (44)	98 f	14.5	14.4	125 f	25.0	24.3	3.40 a,b
17 March 2002 (76)	70 g	16.4	15.1	100 g	25.5	24.5	2.65 c
16 April 2002 (106)	54 h	19.6	18.4	82 h	26.5	23.7	2.25 d
18 May 2002 (138)	44 I	24.9	23.0	66 i	29.5	24.9	2.10 d,e
15 September 2002 (258)	232 a	13.2	10.9	271 a	22.0	20.5	1.60 f
14 October 2002 (287)	204 b	11.4	10.3	242 b	22.2	20.4	1.80 e,f
12 November 2002 (316)	180 c	10.2	10.3	214 c	22.9	21.5	2.00 d,e

In each column means with a common letter are not significantly different at 0.05 level.

dt of 0.02 in original model resulted in yield over-prediction in unseasonally early and early sowing dates. F3E model worked better compared to original model at sowing dates of DOY 258, 288 and 319, but its performance was worse for sowing date of DOY 15 (Fig. 5a). In general, implementation of Fig. 3 function in original model (i.e., F3E model) improved the performance of original model; RMSD decreased and correlation between observed and simulated yields increased (Fig. 5). However, F3E model performance was not satisfactory. Using modified model resulted in yield prediction close to measured yield at all sowing dates with RMSD of 0.28 t ha⁻¹ (11% of observed mean) and correlation coefficient of 0.93

between observed and simulated yields, indicating better performance compared to original and F3E models (Fig. 5). With both original and F3E models, unreal grain yield predicted for sowing date DOY 226. The problem of unreal yield prediction for sowing date DOY 226 was solved with the modified model (Fig. 5). Therefore, a simple modification applied in the modified model well accounted for temperature effects on dHI/dt and could improve the model performance significantly.

In order to indicate how the modified model avoids unreal yield prediction for sowing date DOY 226 and how the model achieves good performance for sowing dates of DOY 258 through DOY 15, we compared

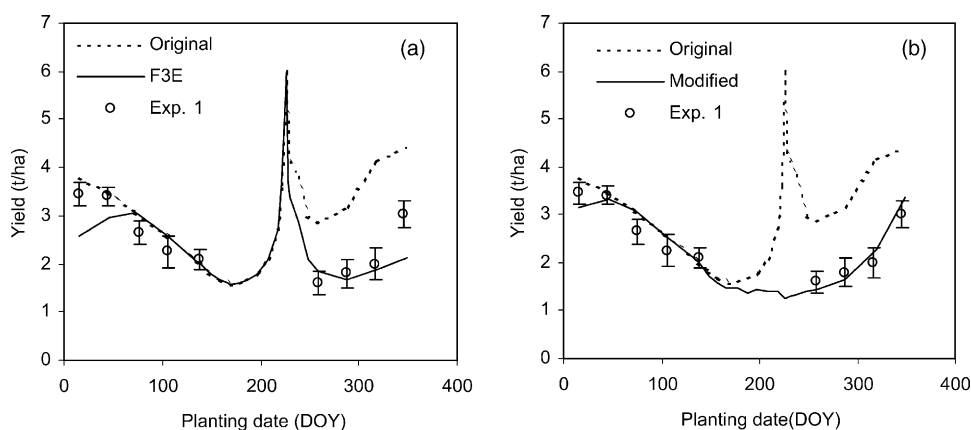


Fig. 5. Observed chickpea grain yields at different sowing dates at Gorgan (exp. 1) and the simulated yields by original, F3E and modified models. Vertical bars indicate standard deviation for observed yield. Root mean square of deviations (RMSD) are 1.14 t ha⁻¹ for original model, 0.52 t ha⁻¹ for F3E model and 0.28 t ha⁻¹ for modified model and the correlation coefficient between simulated and observed yields are 0.46 for original model, 0.70 for F3E model and 0.93 for modified model.

Table 3

Some characteristics of chickpea crops sown on 14 August (DOY 226) 2002 and 12 December (DOY 346) 2001 in Gorgan simulated by original, F3E and modified model

Characteristic	14 August	12 December
Pre-seed growth phase		
Duration (days)	94	166
Mean temperature (°C)	24.2	11.4
Mean solar radiation (MJ m ⁻² day ⁻¹)	14.0	12.1
Seed growth phase		
Duration (days)	51	21
Mean temperature (°C)	8.4	23.7
Mean solar radiation (MJ m ⁻² day ⁻¹)	8.3	22.0
Total biomass (t ha ⁻¹)	6.95	12.45
Grain yield (t ha ⁻¹)		
Original model	6.00	4.43
F3E model	6.00	2.07
Modified model	1.27	3.27
Harvest index (%)		
Original model	75	31
F3E model	75	14
Modified model	16	23

simulation results for sowing dates of DOY 226 and DOY 346 (Table 3). At sowing date of DOY 226, the crop has a pre-seed growth period of 94 days which occurs during mid-August through November with mean temperature of 24 °C and solar radiation of 14 MJ m⁻² day⁻¹. However, seed growth period of the crop coincides with low temperature (8.4 °C) and solar radiation (8.3 MJ m⁻² day⁻¹) of December and

January, which are not suitable for seed growth. Original and F3E models use a dHI/dt of 0.02 to calculate daily seed growth. For F3E model this value is calculated based on the equation presented in Fig. 3. Using this value of dHI/dt during a long period of seed growth (51 days) resulted in unreal yield and HI predictions of 6.00 t ha⁻¹ and 75%, respectively, by the original and F3E models. Therefore, the correction of dHI/dt only for pre-seed growth conditions is not adequate and dHI/dt should also be adjusted for environmental conditions during seed growth period. However, modified model, which limits seed growth to dry matter production during seed growth period and remobilization, predicts grain yield of 1.27 t ha⁻¹. While there is no measured data for sowing date of DOY 226, high yields for crops sown around this time of year are unrealistic based on local experience, but grain yield of around 1.27 t ha⁻¹ by modified model seems acceptable.

At sowing date of DOY 346, the crop has a long pre-seed growth of 170 days from mid-December through May. This long pre-seed growth period results in excessive vegetative growth as indicated by total biomass of 12.45 t ha⁻¹ (Table 3). Thus, using a dHI/dt of 0.02 in original model resulted in 47% over-prediction of grain yield (4.43 t ha⁻¹ versus 3.02 t ha⁻¹). F3E model used a dHI/dt of 0.012 calculated using mean temperature of 11.4 °C during pre-seed growth phase. This value of dHI/dt in combination of the predicted seed growth period of 21 days resulted in 32% under-estimation of grain yield (2.07 t ha⁻¹ versus 3.02 t ha⁻¹). However, source-limited HI-based approach implemented in modified

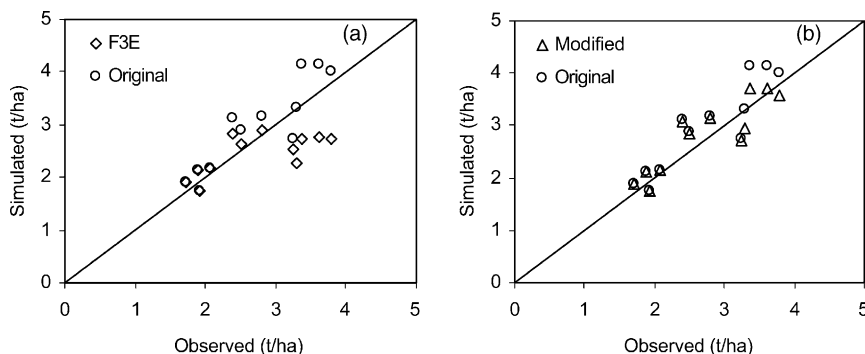


Fig. 6. Simulated chickpea yield for exp. 2 by original, F3E and modified models vs. the observed yields. Root mean square of deviations (RMSD) are 0.44 t ha⁻¹ for original model, 0.47 t ha⁻¹ for F3E model and 0.36 t ha⁻¹ for modified model and the correlation coefficient between simulated and observed yields are 0.91 for original model, 0.79 for F3E model and 0.88 for modified model.

model lead to grain yield prediction of 3.27 t ha^{-1} , which is only 8% greater than measured yield. Mean temperature of 24°C and solar radiation of $22 \text{ MJ m}^{-2} \text{ day}^{-1}$ during seed growth period are favorable for dry matter production.

Simulation of exp. 2 led to similar results (Fig. 6). Mean observed yields were 3.53 t ha^{-1} for sowing date of 6 January, 2.74 t ha^{-1} for sowing date of 6 March and 1.92 t ha^{-1} for sowing date of 28 April. Days to flowering (maturity) for the mentioned sowing dates were 127 (175), 75 (124) and 48 (86), respectively. Original model predicted similar yields to observed yields for sowing date of DOY 118, but at sowing dates of DOY 5 and 65, original model over-predicted grain yield for plant densities of 30, 45 and 60 plant m^{-2} (Fig. 6). F3E model predicted similar yields to observed yields and yields simulated by original model for sowing date of 118. At sowing date of DOY 65, yield predictions of F3E model were closer to observed yields, but at sowing date of DOY 5, grain yields were under-estimated by the model (Fig. 6a). Yield over-prediction by original model and yield under-estimation by F3E model for sowing date of DOY 5 were in agreement with the result of simulation of the exp. 1. Modified model showed better performance for all sowing dates with lower RMSD (Fig. 6b).

3.3. Evaluation of the modified model

Soltani et al. (1999) evaluated their chickpea model for rainfed and irrigated conditions of Iran. Data set used for the model evaluation were mainly from arid and semiarid conditions of Iran. A few points were related to Gorgan, but sowing date of DOY 85. A brief description of the nature of the water deficit stress

under rainfed conditions of the chickpea growing areas of Iran will be helpful. Main chickpea production in Iran is obtained under arid and semiarid conditions. Under these conditions, terminal drought is a widely occurred pattern of soil water availability in which drought is terminal following the major growth period during which water availability is adequate. Soltani et al. (2000b) simulated dynamics of fraction of transpirable soil water during chickpea growing season in a year with the highest simulated yield and in a year with the lowest simulated yield under rainfed conditions of Tabriz and Gorgan (Fig. 7). The FTSW lower than 0.4 can adversely affect photosynthesis of chickpea (Soltani et al., 1999, 2000b). Monthly climatic data of Tabriz (with cold semiarid climate) and Gorgan (with temperate sub-humid climate) are presented in Soltani et al. (1999). Tabriz has cold winters so chickpea is sown in spring, but Gorgan has mild winters and chickpea is sown in autumn. Averages of rainfall (temperature) during chickpea growing period (April–July in Tabriz and December–May in Gorgan) are 120 mm (19°C) and 300 mm (12°C) for Tabriz and Gorgan, respectively. In Tabriz crop experiences a typical terminal drought stress with a long duration (30–40% of the growing season) and high intensity, but in Gorgan this terminal drought stress is short duration and intermittent with low intensity (Fig. 7).

In another study, Soltani et al. (2001b) evaluated the time when terminal drought stress initiated in Maragheh (near Tabriz), a typical chickpea production area in the north west of Iran with a cold semiarid climate. They showed that during the 1/2 to 2/3 of growing season, the crop does not experience drought stress seriously, but thereafter, the crop experiences an increasing water deficit stress. They concluded that in

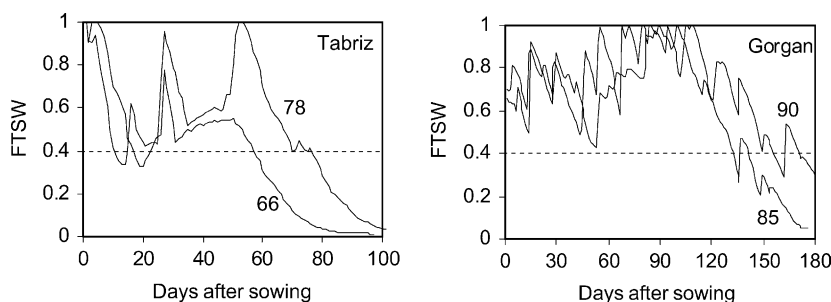


Fig. 7. Seasonal change of FTSW in the root zone (not soil profile) in years with the highest simulated yield (1978 for Tabriz and 1990 for Gorgan) and with the lowest simulated yield (1966 for Tabriz and 1985 for Gorgan) (redrawn from Soltani et al. (2000b)).

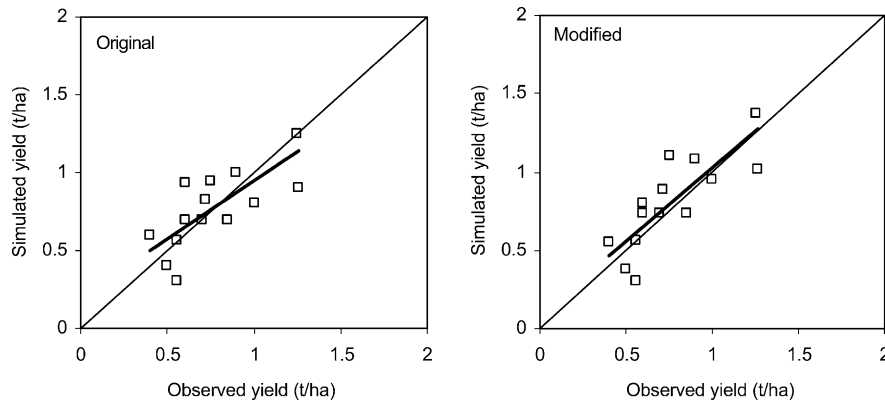


Fig. 8. Simulated yield using original and modified models vs. observed yield (cv. Jam and Kaka) based on independent data sets under water-limited (rainfed) conditions. The solid line is 1:1 line and the thick line is the regression line. Root mean square of deviations (RMSD) are 0.19 t ha^{-1} for original model and 0.18 t ha^{-1} for modified model and the correlation coefficient between simulated and observed yields are 0.77 for original model and 0.84 for modified model.

100% of years terminal drought stress is occurred at a time between flowering and beginning seed growth.

Under above described conditions, crop yield is highly source-limited because there is no serious pre-flowering water deficit. Thus, it is expected that the modified model improve the prediction of grain yield.

Re-evaluation of chickpea model of Soltani et al. (1999) that included the modified HI-based approach and its comparison to original model under water-limited and irrigated conditions are presented in Figs. 8 and 9. Original model accounted for 60% of yield variation under rainfed conditions. RMSD of the

original model was 0.19 t ha^{-1} that is 24% of the observed mean yield (0.79 t ha^{-1}). Using modified model, under exactly the same conditions, declined RMSD by 4% and improved correlation between simulated and observed yield by 9%. As mentioned earlier, the modified model does not have any algorithms for cessation of HI increase under severe drought conditions. Thus, modified HI-based approach has well accounted for drought effects on dHI/dt .

Under irrigated conditions, original model accounted for 67% of the yield variability with RMSD of 0.29 t ha^{-1} that was 12% of the observed mean

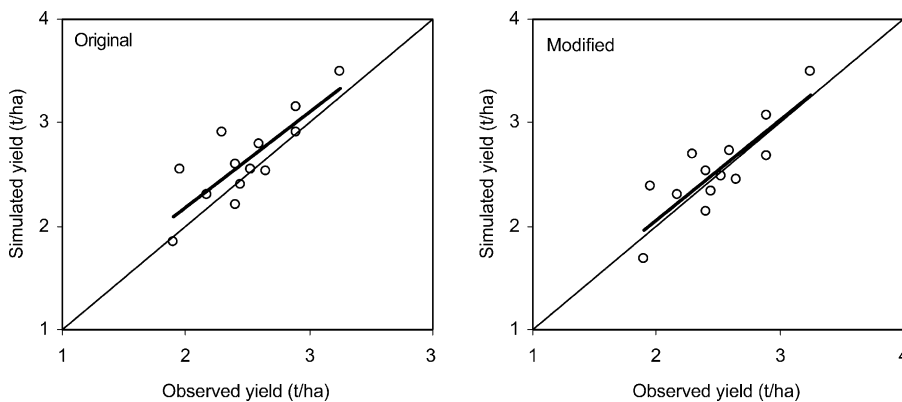


Fig. 9. Simulated yield using original and modified models vs. observed yield (cv. Jam and Kaka) based on independent data sets under well-watered (irrigated) conditions. The solid line is 1:1 line and the thick line is the regression line. Root mean square of deviations (RMSD) are 0.29 t ha^{-1} for original model and 0.24 t ha^{-1} for modified model and the correlation coefficient between simulated and observed yields are 0.82 for original model and 0.84 for modified model.

yield (2.49 t ha^{-1}). Original model over-predicted grain yields by 142 kg ha^{-1} in average. With modified model RMSD decreased by 16% and correlation between simulated and observed yield by 2%. Average yield over-prediction declined to 44 kg ha^{-1} . It seems that the modified HI-based approach has not greatly improved model performance under these conditions, but the improvement is not negligible. The reasons of the improvement are similar to those discussed in Section 3.2.

4. Conclusion

Overall, the field experiments showed that dHI/dt is not constant in chickpea under diverse growth conditions. However, this does not mean that the HI-based approach is deficient for yield prediction. Using a simple modification in the HI approach based on supply of assimilate (source-limited HI approach) could greatly improve model performance under special conditions of unseasonally early and early sowings at Gorgan. This modification well accounted for temperature and drought effects on HI dynamics and resulted in lower RMSD and higher correlation between simulated and observed yields under conditions of major chickpea producing areas of Iran. Therefore, we recommend that the modification to be applied in the other HI-based models.

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